Sumon Nath

Offensive Security Engineer | Penetration Testing (VAPT) Lead | Red Team Operator

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PROFESSIONAL SUMMARY

Offensive security engineer with 3+ years and 100+ penetration tests delivered across banking and financial services (BFSI), fintech, government and enterprise environments. Microsoft Most Valuable Security Researcher 2022 and 2023, five assigned Common Vulnerabilities and Exposures (CVE) identifiers, and vulnerabilities accepted by Microsoft, Google, Apple and NASA. Full lifecycle Vulnerability Assessment and Penetration Testing (VAPT), MITRE ATT&CK red team operations, secure code review and cloud security engineering.

KEY ACHIEVEMENTS

- Microsoft Most Valuable Security Researcher (MVR), 2022 and 2023 — awarded for high-impact vulnerability submissions to Microsoft products.
- Five CVE identifiers assigned through coordinated vendor disclosure, plus a contribution to a MITRE Common Weakness Enumeration (CWE) entry.
- Vulnerabilities accepted by Microsoft, Google, Apple and NASA across 100+ public and private bug bounty programs.
- First-author peer-reviewed book chapter published by Springer Nature (879 accesses as of November 2026).

WORK EXPERIENCE

Senior Security Researcher, Team Lead

Alphawave Technology Pvt Ltd (Mitigata Security) | Bengaluru, India | Jan 2026 - Present

- Delivered penetration testing across 100+ enterprise clients covering web, REST/GraphQL/SOAP APIs, mobile (iOS and Android), internal and external network, Active Directory and perimeter firewall scope, holding sole technical sign-off on every report released.
- Cut time-to-first-finding on every engagement by engineering a reconnaissance and attack-surface automation pipeline (Amass, httpx, nuclei) that performs continuous asset enumeration and vulnerability chaining in place of manual discovery.
- Executed MITRE ATT&CK-aligned adversary emulation end to end: stood up command-and-control (C2) infrastructure with redirectors and domain fronting, defeated antivirus and Endpoint Detection and Response (EDR) through AMSI/ETW patching and reflective loading, then chained Kerberos abuse, credential harvesting and lateral movement to controlled exfiltration.
- Restored testing coverage against EDR-protected estates by writing custom offensive tooling, payloads and loaders in Python, C and Go where signatured commercial tooling was blocked.

Senior Software Engineer and Security Researcher

Loginsoft Pvt Ltd | Hyderabad, India | May 2024 - Nov 2025

- Gave detection engineering a reproducible ground truth by reverse-engineering disclosed CVEs into working proof-of-concept exploits and shipping them as deterministic test cases in instrumented sandbox environments.
- Reduced false positives in SAST and SCA tooling (Checkmarx, Snyk) by authoring custom query rules with purpose-built sink and source definitions, widening true-positive detection coverage at the same time.

Security Intern

Centre of Excellence in Cyber Security, VIT Bhopal | Bhopal, India | Jan 2023 - Feb 2024

- Designed and operated 20+ Capture The Flag (CTF) events in Jeopardy and Attack-Defense formats for 1,000+ participants, authoring challenges in web exploitation, binary exploitation, reverse engineering, cryptography and forensics, sponsored by Offensive Security, Bugcrowd and HackerOne.

TECHNICAL SKILLS

Penetration Testing and Red Teaming: Vulnerability Assessment and Penetration Testing (VAPT), Web Application Penetration Testing, API Security Testing (REST, GraphQL, SOAP), Mobile Application Penetration Testing (iOS, Android), Network Penetration Testing, Internal and External Infrastructure Testing, Active Directory Security, Kerberos Attacks, Red Team Operations, Adversary Emulation

Vulnerability Research and Reverse Engineering: Vulnerability Research, Reverse Engineering, Binary Exploitation, Exploit Development, CVE Research and Reproduction, Proof of Concept (PoC) Development, Coordinated Vulnerability Disclosure, Bug Bounty Hunting, Offensive Tooling Development, Cryptography

Application and Cloud Security: Secure Code Review, Static Application Security Testing (SAST), Dynamic Application Security Testing (DAST), Software Composition Analysis (SCA), Software Bill of Materials (SBOM), Supply Chain Security, Threat Modeling (STRIDE), Secure Software Development Lifecycle (SDLC), DevSecOps and CI/CD Security, Cloud Security Posture Management (CSPM)

Tools: Burp Suite, Metasploit, Cobalt Strike, Nmap, Nessus, nuclei, Amass, httpx, MobSF, Ghidra

Programming and Scripting: Python, C, C++, Go, Rust, PowerShell, Bash, JavaScript, TypeScript, SQL

KEY PROJECTS

CtrlSec - AI-Powered Autonomous Security Operations Platform

ctrlsec.io

- Architected and built a full-stack autonomous threat-detection platform single-handedly — backend services, Large Language Model (LLM) inference pipeline, threat-logic layer, compliance engine and frontend console — across Python, Go, Node.js, React and Next.js.

AI-SAST Scanner - Deep-Learning Vulnerability Detection

- Built a deep-learning static analysis engine in PyTorch (Convolutional Neural Network with Bidirectional Long Short-Term Memory and Transformer attention) performing cross-function taint analysis over 35 vulnerability classes across 9 programming languages.

PUBLICATIONS

- Nath, S. et al. (2025). Blockchain-Based Knowledge Repository for Training Artificial Intelligence Models: Bridging AIML with Decentralized Data. Studies in Systems, Decision and Control, vol. 587. Springer Nature Switzerland. DOI: 10.1007/978-3-031-87584-7_126

EDUCATION AND CERTIFICATIONS

Integrated Master of Technology (M.Tech), Computer Science and Engineering

Specialisation in Cybersecurity

Vellore Institute of Technology (VIT), Bhopal, India | 2020 - 2025

- Certified Ethical Hacker (CEH) - EC-Council
- Offensive Security Certified Professional (OSCP) - In Progress